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# Self-adaptive LLM instructions optimization for aspect-based sentiment analysis by incorporating emotion-oriented in-contexts

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Aspect-based Sentiment Analysis (ABSA) is a vital NLP task that identifies sentiment towards specific entities or aspect terms within a text. Recently, large language models (LLMs) have shown impressive capabilities in semantic comprehension and logical inference. However, LLM hallucinations pose challenges in accurately determining sentiment polarity for aspect terms, leading to performance issues. Moreover, current ABSA methods often fail to fully leverage the vast prior knowledge embedded within LLMs, resulting in suboptimal classification outcomes for specific aspects. Inspired by these challenges, we propose the BYD-OBS-ABSA framework—'Beyond Simple Observations, Embracing Comprehensive Contextual Insights' for ABSA tasks. This framework leverages unique in-context constraints, backgrounds and analogical reasoning to address LLM hallucinations and uses self-adaptive bootstrap instructions optimization to enhance LLM predictions. BYD-OBS-ABSA integrates various in-context augmentation strategies, including emotion-oriented backgrounds, constraints, and analogical reasoning. BYD-OBS-ABSA further improves initial LLM instructions through adaptive iterative optimization using a random search bootstrap algorithm, maximizing the benefits of LLM prompting. Extensive zero/few-shot experiments with `GPT-3.5-turbo` across six public datasets validate the effectiveness and robustness of our framework, even surpassing human judgment in certain scenarios.

**KEYWORDS**
Aspect-based Sentiment Analysis, LLM Prompting Engineering, In-context Learning, Symbolic Chain-of-Thought, In-context Constraint, Analogical Reasoning, Bootstrap Optimization.

**Abbreviations:** XXX.

[*]Weiqiang Jin and Junli Wang contributed equally to this work. Biao Zhao and Guang Yang are the corresponding authors.

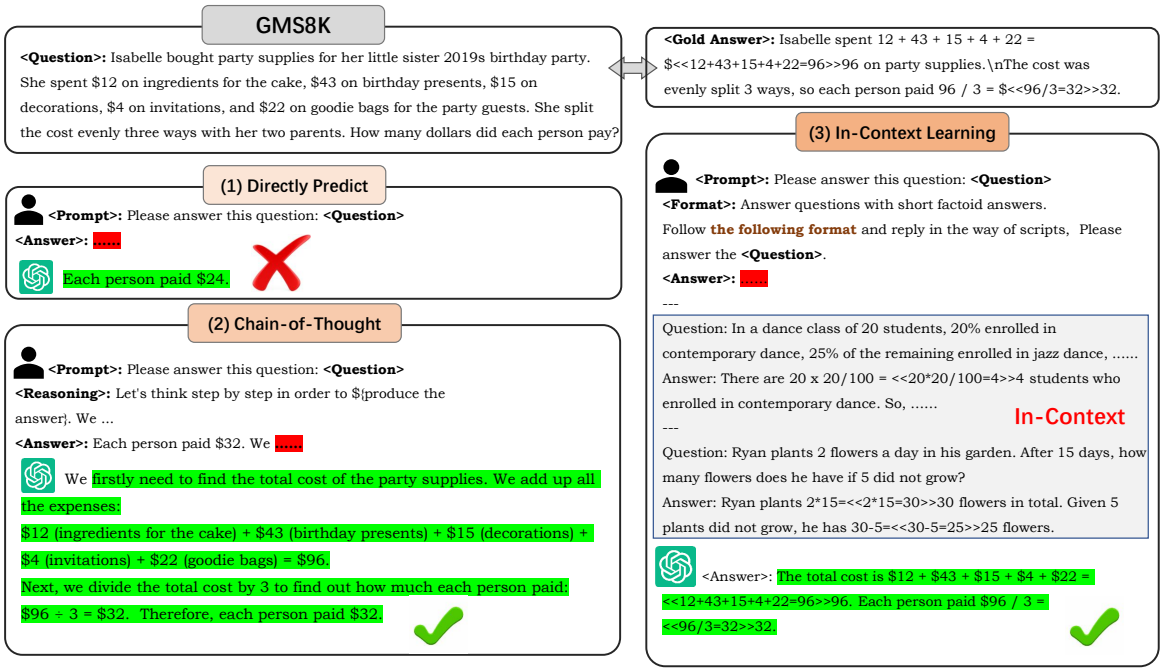



**FIGURE 1** Three LLM prompting methods and their query interaction effects on a multi-step reasoning elementary math problem set **GSM8K**: directly predict (top left), Chain-of-Thought (bottom left), In-Context Learning (right). The example shown is about isabelle splitting the cost of party supplies for her sister's birthday party.

## 1 | INTRODUCTION

Aspect-based Sentiment Analysis (ABSA) [1, 2, 3] is a crucial task in natural language processing (NLP). It involves identifying sentiment polarities directed towards specific aspects or features mentioned in a text. For example, in a restaurant review, ABSA would not only determine the overall sentiment of the review but also analyze sentiments towards specific aspects such as food quality, service, and ambiance. Its significance lies in its ability to provide fine-grained insights into customer opinions, enhancing decision-making in areas such as marketing, product development, and customer service.

However, in previous Artificial Intelligence (AI) research, traditional deep learning (DL)-based ABSA approaches primarily relied on pretrained word embeddings and fine-tuning small language models (SLMs) like BERT, RoBERTa, and XLNET [4, 5, 6]. These approaches faced several challenges and limitations [1, 2, 3]. Requiring large amounts of domain-specific training data, which can be difficult and costly to obtain. Despite deep learning reduces the need for manual feature engineering, traditional models still require extensive preprocessing and feature engineering to enhance performance. Poor domain transferability makes it challenging to apply a model trained in one domain to another, and it also struggles to handle multilingual environments and cross-language tasks. The relatively small parameter size of these models causes performance bottlenecks, limiting their effectiveness in capturing nuanced sentiments. Overall, these aforementioned challenges underscore the necessity for more robust and scalable solutions.

Overcoming these obstacles requires the adoption of more powerful techniques, such as recently proposed large language models (LLMs), to further enhance the effectiveness and push the upper limits of ABSA. The emergence of various LLMs, such as Llama [7], Gemini [8], Claude [9], and especially OpenAI's ChatGPT [10, 11, 12], has sparked widespread speculation in the press about their potential to disrupt society, economy, and even the political landscape. Their astonishing language abilities have generated strong curiosity among scholars regarding their performance in various domains.

Prompting techniques, by equipping LLMs with rich vocabulary, professional expression, and advanced response mechanism, enable machines to better understand human intentions and communicate more effectively. These techniques have become a core technology, evolving in tandem with LLMs. Recent advancements in prompting techniques, such as Chain-of-Thought (CoT) [13, 14], In-Context Learning (ICL) [15, 16, 17], and Reasoning and Acting (ReAct) [18], have shown significant improvements in handling complex tasks. CoT prompts LLMs to engage in step-by-step reasoning, making the decision-making process more transparent and robust, while ICL provides context through examples, enhancing model understanding without needing gradient updates. ReAct combines reasoning with actionable steps, allowing models to interact more dynamically with tasks. These prompting techniques highlight the importance of designing effective prompts to fully leverage the capabilities of LLMs.

Figure 1 demonstrates the performance and decision-making process of the prompting techniques: directly predict, Chain-

of-Thought, and In-Context Learning, on **GSM8K Math Problems** in eliciting LLMs to handle complex queries. **GSM8K** [19] is a linguistically diverse dataset of grade school math problems designed to evaluate and improve language models' multi-step reasoning abilities. The entire interaction uses LLM `GPT-3.5-turbo` as the core engine. Through observation, we find that for some complex problems, Chain-of-Thought and In-Context Learning can better decompose problem steps, regularize the reasoning process, and standardize the response format, addressing the complex logical reasoning issues that directly predict cannot solve.

Although representative prompting techniques positively impact the elicitation of LLMs' response potential, their improvements are limited, particularly in specific domains and downstream linguistic applications. For instance, despite Chain-of-Thought (CoT) prompting facilitating step-by-step reasoning for more robust decision-making, it remains a static and opaque process. This uncontrollability in the reasoning chain can lead to issues such as factual hallucinations and error propagation due to reliance on internally generated ideas. Similarly, In-Context Learning (ICL) involves concatenating input-label pairs from the training data and presenting similar query cases in a list format. The efficacy of LLM predictions in ICL heavily depends on the quality of annotated input-label pairs. Consequently, the performance degrades significantly when these labels are randomized or inaccurately annotated, highlighting the need for more robust and context-aware prompting strategies.

Another critical challenge in current ABSA works [2, 20, 1] is the insufficient extraction of prior knowledge embedded within LLMs, leading to subpar classification results for specific aspects. This underutilization of LLMs' extensive prior knowledge hampers the performance of aspect-specific sentiment analysis, necessitating strategies that can more comprehensively tap into the wealth of information these models possess.

In human communication, contextual and emotional subtleties are integral to accurately interpreting sentiments. Humans leverage these contextual cues and emotional nuances to discern and evaluate sentiments towards specific entities within a text. In more comprehensive contexts, human judgment excels in assessing and analyzing opinions towards specified entities due to its ability to grasp nuanced meanings, understand contextual subtleties, and interpret emotional cues. Humans can integrate broad situational factors, cultural references, and historical context into their evaluations, allowing for a deeper and more accurate sentiment analysis.

Similarly, emotion-oriented in-contexts in NLP tasks are similarly pivotal for improving the precision and comprehensiveness of reasoning in fine-grained linguistic tasks, particularly in ABSA. Based on this understanding, we propose that incorporating emotion-oriented contexts into LLMs can enhance their interpretative and reasoning capabilities, leading to significantly improved response quality and accuracy. Aligning LLMs' analytical processes with human-like emotional and contextual understanding potentially addresses limitations in previous sentiment analysis methodologies.

Recognizing the crucial role of emotion-oriented in-contexts for ABSA tasks, we have worked to improve these contextual factors and integrate them into the prompting process to offer more effective instructional guidance for LLMs. Building on the integration of Chain-of-Thought and In-Context Learning, we systematically optimize the LLMs' query process and strengthen response quality through the construction and utilization of emotion-oriented in-contexts. To this end, we introduce a novel LLM prompting-based ABSA framework, called '*Beyond Simple Observations, Embracing Comprehensive Contextual Insights*' (BYD-OBS-ABSA). This innovative framework seeks to bolster LLMs' interpretative and reasoning abilities, aligning their analytical processes more closely with human-like emotional and contextual understanding, thereby effectively addressing the limitations of previous sentiment analysis methodologies.

Specifically, in the prompting procedure of the BYD-OBS-ABSA framework, we have comprehensively considered emotion-oriented in-contexts from three different perspectives: in-context constraints, in-context backgrounds, and in-context analogies. Each in-context factor has been implemented, obtained and infused into the LLM's prompting via its corresponding in-context acquisition/infusion strategy.

**Below are the details of the infusions of emotion-oriented in-context constraints, backgrounds, and analogies:**

**Emotion-oriented in-context constraint:** In the specific implementation of in-context constraints, we begin by constructing variant queries that hypothesize different polarities for the aspect term within the current sentence. Specifically, we generate three variant queries and ask the LLM whether each hypothesis is correct or incorrect (i.e., Yes/ No), alongside the original query. Then, through a consistency alignment mechanism, we perform a secondary evaluation of these intermediate responses based on LLMs. All these procedures are driven by the Chain-of-Thought prompting engineering [13, 14], which directs the LLM through a systematic reasoning process. Additionally, we employ symbolic representations (Symbolic Premises), such as first-order logic, to ensure the consistency and accuracy of the query mapping rules. By combining this with Chain-of-Thought, referred to as Symbolic Chain-of-Thought (SymbCoT), we reduce ambiguity and enhance the overall reliability of the reasoning process.

**Emotion-oriented in-context backgrounds:** The integration of external knowledge regarding the target entity or aspect term significantly enhances the efficacy of emotion-oriented in-context techniques in our work [21]. To integrate aspect term's background information into sentiment identification, we have the LLMs automatically generate a brief introduction for each aspect term. This introduction, termed in-context backgrounds, is then used as an additional prompt to enhance the final sentiment determination. This process ensures a more nuanced and context-aware sentiment analysis by leveraging the LLM's ability to provide background information.

**Emotion-oriented in-context analogies:** In Artificial Intelligence (AI), teaching computers to perform complex reasoning like humans has always been a significant challenge. Current prompting techniques, such as the CoT and ICL, often require annotated examples of the reasoning process as input, which is labor-intensive and may have limited relevance to the target problem. Is there a way for language models to generate the most relevant problem-solving examples themselves to better address the issue at hand? Thus, we devised emotion-oriented in-context analogies in our BYD-OBS-ABSA framework, much like how humans use analogical reasoning to derive solutions based on past problem-solving experiences. This technology involves having the LLMs automatically generate analogical examples that are emotionally and contextually relevant to the given problem, and then guiding the LLM in solving the problem by leveraging similar past experiences or scenarios.

Our contributions in this paper can be summarized as follows:

- We systematically studied the shortcomings and limitations of current mainstream LLM prompting methods, including CoT and ICL. To solve the challenges of LLM hallucinations and fully utilize the prior knowledge embedded within LLM, we introduced a novel LLM prompting-based ABSA framework, called 'Beyond Simple Observations, Embracing Comprehensive Contextual Insights' (BYD-OBS-ABSA).
- BYD-OBS-ABSA introduces the acquisition and integration of emotion-oriented in-context constraints, backgrounds, and analogical reasoning, which is particularly effective in addressing LLM hallucinations. By employing a variety of infusion strategies, we ensure the normative, comprehensive, and robust handling of the prompting process for LLM-based ABSA tasks.
- We further enhance the performance and robustness of BYD-OBS-ABSA by incorporating *Self-Adaptive Bootstrap Instructions Optimization*. This method fine-tunes the prompts and demonstrations provided to LLMs, ensuring they generate more accurate and reliable sentiment predictions.
- Comprehensive zero/few-shot experiments based on `GPT-3.5-turbo` across six public datasets validate the effectiveness and robustness of BYD-OBS-ABSA, even surpassing human judgment in several certain cases.
- To support further research on LLM-prompting for ABSA, we have made our approach's code available on GitHub[1].

The remainder of this paper is organized as follows. In Section 2, we provide a comprehensive review of related work, focusing on the advancements and methodologies in ABSA and LLMs. Section 3, we formulate the technical concepts in detail, including the formulations of '*Aspect-based Sentiment Analysis*' (ABSA) task and '*LLM-based Prompting Engineering*'. Section 4 introduces our proposed framework, BYD-OBS-ABSA, detailing its design, the integration of emotion-oriented in-context constraints, backgrounds, and analogical reasoning, and how this prompting works. In Section 5, we present the experimental results and compare the performance of BYD-OBS-ABSA with existing approaches. Section 6 offers comprehensive discussions on the implications, limitations, and potential areas for future improvement of our method. Finally, Section 7 concludes the paper and suggests directions for future research.

## 2 | RELATED WORKS

### 2.1 | Aspect-based Sentiment Analysis

Aspect-Based Sentiment Analysis (ABSA) [1, 2, 3] is a significant research field in linguistics and semantics that has garnered increased attention in the era of big data. Gianni et al. [3] provided a taxonomy for ABSA models, categorizing them into knowledge-based, machine learning, and hybrid models. Zhou et al. [1] focused on deep learning-based ABSA, conducting a systematic review and comparing performance. DAMO Academy [2] addressed the lack of systematic reviews on ABSA tasks, introducing a new taxonomy based on sentiment elements and emphasizing the use of PLMs to enhance ABSA across domains and languages.

In the past decades, significant progress [22, 23, 24, 20] has been made in ABSA using SLMs [4, 5, 25, 26]. Alexander et al. [22] applied deep transfer learning to the ASBA task and proposed Domain Adaptation through BERT language model (BERT-ADA). Zeng et al. [23] introduced LCF-ABSA, a novel Local Context Focus mechanism for ABSA, addressing the limitations of traditional methods that primarily consider global context, thus avoiding misjudging sentiment polarities of targeted aspects. Song et al. [24] proposed an Attentional Encoder Network (AEN) for targeted sentiment classification, AEN-GloVe and AEN-BERT, which replaces recurrence with attention-based encoders and introduces label smoothing regularization. Jin et al. [21] integrated dictionary-based explanatory knowledge into LM-based ASBA models, demonstrating the significance of incorporating external common-sense knowledge in sentiment analysis. Xu et al. [27] proposed a task called Review Reading Comprehension (RRC) that transforms customer reviews into a knowledge source to answer user questions, exploring a novel BERT post-training method to enhance BERT's performance in ABSA tasks. Nazir et al. [20] introduced an Interactive Attention Fusion network with Local and Global perspective for ABSA tasks (IAF-LG), which enhances the accuracy of ABSA by simultaneously learning local and global semantics. These SLM-based approaches have achieved significant breakthroughs in ABSA tasks. However, the limited parameter size and small-scale pre-training

corpora of small language models have constrained their performance in various text processing tasks.

With the advancement of larger language models (LLM)-based prompting techniques, many sentiment analysis models [28, 29, 17, 30] based on LLM [7, 8, 9, 10] have been developed. Scaria et al. [31] introduce InstructABSA, an LLM-based instruction learning paradigm for ABSA that incorporates positive, negative, and neutral examples into task-specific prompts, significantly enhancing performance. Simmering et al. [28] assess GPT-3.5 and GPT-4 [12, 11] for ABSA in various settings, finding that fine-tuned GPT-3.5 achieves an F1 score of 83.8 on SemEval-2014 Task 4 [32], surpassing InstructABSA by 5.7% but at a higher inference cost. Zheng et al. [29] a novel instruction tuning method with retrieval-based example ranking for ABSA. This method employs a LM T5 [25] as both a scorer and generator, using an alternating training schema to rank examples and optimize instructions based on log-likelihood. Yang et al. [17] propose FaiMA, a Feature-aware In-Context Learning framework for Multi-domain ABSA. FaiMA uses a Multi-head Graph Attention Network Encoder (MGATE) and In-Context Learning (ICL) [15] to integrate traditional techniques with LLMs, effectively handling linguistic, domain, and sentiment features. Ding et al. [30] introduce LLM-CL, a Continual Learning framework for ABSA that addresses domain knowledge variability and catastrophic forgetting, employing a domain knowledge decoupling with an orthogonal constraint and a domain knowledge warmup strategy.

In summary, future research in ABSA will continue to focus on solutions based on LLMs. Future efforts will continually discover advanced instruction engineering and prompt techniques to enhance model performance and generalization capabilities while reducing computational costs. Optimizing instruction learning and prompt engineering will be crucial, particularly in zero-shot and few-shot settings. Addressing these challenges will be key to leveraging LLMs more effectively in ABSA tasks.

## 2.2 | Large Language Model

Before the advent of the first truly large-scale language model ChatGPT, we witnessed the success of some small pre-trained language models, including BERT, RoBERTa, T5, XLNET, and GPT-2 [4, 5, 25, 6, 10], in various language understanding tasks. Since ChatGPT [26, 10, 11, 12] proposed, Large Language Models (LLMs) have gradually revolutionized AI by leveraging vast data and computational power to achieve unprecedented performance in NLP tasks [33, 34]. Models such as OpenAI's GPT-3 and GPT-4 series [26, 10, 12], Meta AI's Llama 3 [7], Google's Gemini series [8], Anthropic's Claude 3 [9], and Kimi AI's Moonshot-v1 series [35] demonstrate significant potential in applications like code generation, text generation, summarization, translation, logical reasoning, question-answering, and dialogue systems. Chinese LLMs are advancing rapidly and are comparable to international counterparts. Du et al. [36] introduced GLM, a General Language Model based on autoregressive blank infilling. Building on GLM, Zhipu AI and Tsinghua University's KEG Lab released ChatGLM3 [37], an open-source dialogue model with impressive performance. Kimi AI's Moonshot-v1 series [35] can handle extremely long prompts, processing up to 2 million Chinese characters, surpassing other LLMs' limitations.

LLMs have made breakthroughs in in-context and few-shot learning, using advanced prompt engineering techniques like Chain-of-Thought (CoT) [13, 14], Reasoning and Acting (ReAct) [18], and Implicit In-context [16] to enhance performance. DeepMind's FunSearch [38] provided new solutions to classic mathematical problems, surpassing human capabilities. Guo et al. [39] developed a data science analysis system based on LLMs for complex machine learning tasks. Wang et al. [40] surveyed LLM-based autonomous agents, highlighting their potential in achieving human-level intelligence through vast web knowledge.

Research on LLMs highlights both their strengths and weaknesses, as well as potential areas for future improvements [41, 42, 43, 44]. Chen et al. [41] explored the role of prompt engineering in optimizing LLM capabilities. Sahoo et al. [42] surveyed prompt engineering approaches, evaluating strengths, limitations, and key aspects. Chen et al. examined LLM applications in finance, healthcare, and law, highlighting significant improvements in diagnostic methods and financial analytics. Google [43] reassessed LLM capabilities, noting they cannot simulate human cognition, as they predict the next token rather than engage in detailed mental processes. Kambhampati et al. [44] critiqued views on LLM capabilities, balancing between over-optimism and reductionist perspectives.

Overall, LLMs have achieved significant success by leveraging vast amounts of data and computational power to achieve unprecedented performance in natural language processing tasks. In the future, LLMs are expected to continue driving the development of artificial intelligence, expanding application areas, and realizing more intelligent and human-like natural language processing systems.

# 3 | PRELIMINARIES

In this work, we focus on ABSA and LLM-based Prompting Engineering, which are critical components in NLP aimed at enhancing the accuracy and contextual appropriateness of responses generated by LLMs.

## 3.1 | Aspect-based Sentiment Analysis

Aspect-based Sentiment Analysis (ABSA) aims to accurately determine the sentiment associated with specific aspect terms (also referred to as target words) within a given sentence. ABSA can be formulated as follows:

Given an input sentence $S = \{w_i\}_{i=1}^{L} = \{w_1, \ldots, w_i, \ldots, w_L\}$ with $L$ tokens, assume the sentence contains $K$ aspect terms $T_K = \{a_j\}_{j=1}^{K}$. Each aspect term $a_j$ is a text span $a_j = [P_j^{\text{start}}; P_j^{\text{end}}]$ in the sentence $S$, where $P_j^{\text{start}}$ and $P_j^{\text{end}}$ denote the positions of the beginning and ending boundaries of the aspect term, respectively.

For each aspect term $a_j$, the ABSA task aims to infer a corresponding sentiment tag $y_j$ from a predefined sentiment polarity set $Y = \{\text{pos}, \text{neg}, \text{neu}\}$ based on the encoded representations of the contextualized semantics in the sentence $S$.

For example, given the user comment "The battery life of this phone is terrible, but the camera quality is excellent", the task is to identify the sentiment towards the target entities "battery life" (negative sentiment) and "camera quality" (positive sentiment).

## 3.2 | LLM-based Prompting Engineering

LLM-based Prompting Engineering involves the design and optimization of input prompts to enhance the performance of large language models (LLMs) in generating accurate and contextually appropriate responses. This process can be mathematically categorized based on the amount of example data provided: zero-shot and few-shot prompting.

### 3.2.1 | Zero-shot Prompting

In zero-shot prompting, the model is provided with a problem statement $x$ without any additional examples or context. The prompt $\varphi$ directly yields $x$: $\varphi(x) = [x]$ This technique relies on the model's pre-existing knowledge to generate a solution $\hat{y}$: $\hat{y} = \text{LLM}(\varphi(x))$. For example, given a task to identify the sentiment in a review, a zero-shot prompt might simply present the review text to the LLM, expecting it to infer the sentiment without further guidance.

Additionally, the zero-shot prompting approach combined with Chain-of-Thought (CoT) enhances the model's performance by instructing it to think step-by-step. This is achieved by appending a general instruction to the problem statement $x$, resulting in $\varphi(x) = [x]$, think step by step. This approach encourages the model to generate intermediate reasoning steps before arriving at the final answer, thereby improving the accuracy and explainability of the response.

### 3.2.2 | Few-shot Prompting

Few-shot prompting provides the LLM with several labeled exemplars to guide its understanding of the task. The method $\varphi$ supplements the problem statement $x$ with $K$ labeled examples, each consisting of a problem statement $x_i$, reasoning steps $r_i$, and a final answer $a_i$: $\varphi(x) = [x_1][r_1][a_1] \ldots [x_K][r_K][a_K][x]$. This approach allows the LLM to learn from the examples and apply similar reasoning to the new problem statement $x$: $\hat{y} = \text{LLM}(\varphi(x))$.

For example, given a task to identify the sentiment in a review, a few-shot prompt might include several examples of sentences with their corresponding aspect terms and sentiment labels. For instance, provided examples might be:

```
#Given:
"The service was excellent" (aspect: service, sentiment: positive),
"The food was bland" (aspect: food, sentiment: negative),
"The ambiance was cozy" (aspect: ambiance, sentiment: positive)

#Following the above examples, please answer the question:
"The staff were friendly but the room was noisy," (aspect: {$aspect},
sentiment: {$sentiment})
```

Given these examples as context, the LLM can better infer the sentiment for the new sentence, "The staff were friendly but the room was noisy," identifying positive sentiment towards "staff" and negative sentiment towards "room".

For few-shot prompting, adding Chain-of-Thought (CoT) also involves appending a general instruction like "step by step" to the problem statement, resulting in: $\varphi(x) = [x_1][r_1][a_1] \ldots [x_K][r_K][a_K][x]$, *think step by step*. This helps the model generate intermediate reasoning steps, improving accuracy and explainability.

## 4 | OUR FRAMEWORK: BYD-OBS-ABSA

Inspired by the human ability to grasp nuanced meanings and interpret emotional and contextual subtleties to better assist self cognitive judgment, we propose incorporating emotion-oriented in-contexts into LLMs prompting for ABSA. Thus, we introduce BYD-OBS-ABSA, i.e., 'Beyond Simple Observations, Embracing Comprehensive Contextual Insights', a novel LLM-based prompting ABSA framework. It is designed to enhance the capabilities of LLMs, the accuracy, and the depth in determining sentiment polarity for aspect terms within text by leveraging the sophisticated understanding that humans apply in evaluating specific entities within text.

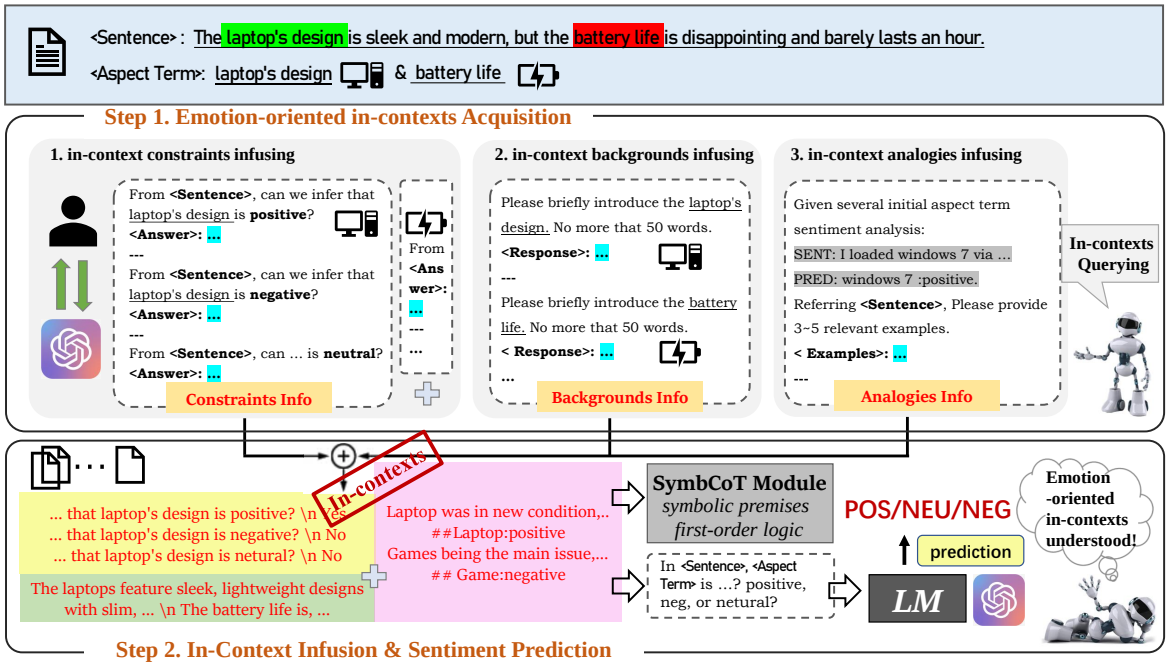



**FIGURE 2** The overall structural illustration of the proposed BYD-OBS-ABSA framework uses an example from **SemEval2014**[32] to explain. We divided the integration of emotion-oriented in-contexts with LLM prompting into two stages: ***Top:*** *Emotion-oriented in-contexts acquisition*: We obtain emotional in-contexts related to the current ABSA problem through a series of queries to the LLM, which includes emotion-oriented in-context backgrounds, constraints, and analogies; ***Bottom:*** *In-contexts infusion & sentiment prediction*: Based on the in-contexts obtained from the previous stage, In-Context Learning is used to provide these contexts for the original ABSA, assisting the LLM in better understanding and predicting sentiment polarity.

### 4.1 | Main Structure

Our approach incorporates emotion-oriented in-context constraints, backgrounds, and analogical reasoning to address the challenges posed by LLM hallucinations and improve overall sentiment analysis performance. As shown in Figure 2, BYD-OBS-ABSA systematically optimizes LLM instructions through two primary stages: **Emotion-oriented in-contexts acquisition** (Section 4.2), and **In-contexts infusion & sentiment prediction** (Section 4.3).

In the first stage, we engage the LLM in multiple rounds of queries to enable it to self-generate relevant contextual information and knowledge. Initially, we create multiple linguistic variants and interact with the LLM to gather its feedback. This feedback, combined with a consistency alignment mechanism, forms the in-context constraints. Through further interactions, we provide brief descriptions of the aspect terms to establish contextual backgrounds. Finally, we guide the LLM to autonomously generate relevant analogies, enhancing its sentiment-oriented understanding.

Corresponding to Figure 2, for the sentence $S$ := "The laptop's design is sleek and modern, but the battery life is disappointing and barely lasts an hour." and the aspect terms $T$ := "laptop's design" and "battery life", we design and obtain responses to the linguistic variants of the query "`From` $S$`, can we infer that the` $T$ `is positive/ negative/ neutral?`". Additionally, the background knowledge for the aspect terms $T$ and relevant analogies/ exemplars towards the sentence $S$ are self-generated and provide by the interactions with LLM.

In the second stage, these obtained in-contexts are utilized to assist the LLM in making accurate sentiment predictions. This

is achieved by leveraging the In-Context Learning (ICL) technique. Returning to the example in Figure 2, we integrate the emotion-oriented in-contexts obtained in the first stage for the original sentence $S$ and aspect terms $T$ using the Symbolic Chain-of-Thought and In-Context Learning techniques. The Symbolic Chain-of-Thought (SymbCoT) module, is used to post-process the in-context constraints information within a consistency alignment mechanism to enhance reasoning and prediction accuracy. Then, through In-Context Learning, these in-contexts serve as a comprehensive background for sentiment analysis of the original sentence. With this enriched context, the LLM can make more accurate and nuanced sentiment judgments.

Note that during the prompting instruction constructions of BYD-OBS-ABSA, using '#' symbols in prompts (e.g., '# Relevant Sentiment-Contained Sentence with entities') improves the structure of responses for LLMs. To ensure the simplicity of Figure 2, the specific prompts in actual implementation might differ slightly from what is shown. Please refer to the subsequent contents for detailed explanations.

Next, we introduce the specific prompting designs for each module of our BYD-OBS-ABSA framework.

## 4.2 | Emotion-Oriented In-Contexts Acquisition

Here, we elaborate on the implementation details of the first stage of BYD-OBS-ABSA, **Emotion-oriented in-contexts acquisition**. This involves describing the inputs and outputs of each module in this stage, as well as the LLM prompting.

### 4.2.1 | In-context Constraints Infusion Strategy

Recent LLM-based prompting engineering explorations [14, 13] have demonstrated the emergent capability to perform multi-step reasoning for tasks involving arithmetic, commonsense, and symbolic reasoning, specifically through the emerging Chain-of-Thought (CoT) technique. Although CoT encourages LLMs to think in step-by-step human-like reasoning, enabling more robust decision-making, a single CoT remains a static black box. The rationality of the model's reasoning chain is uncontrollable, as the entire reasoning process relies on internally generated ideas. This can lead to issues like factual hallucinations and error propagation.

Similarly, in fine-grained classification tasks like ABSA, Reading Comprehension, and Relation Extraction (RE), LLMs' classification results can be influenced by various factors, leading to misjudgments, especially in contexts with less apparent semantic features. Therefore, it is necessary to design linguistic/semantic-level constraints to ensure LLMs systematically complete ABSA tasks. Inspired by this, we propose the emotion-oriented in-context constraints. Specifically, we designed multiple linguistic variants of the original sentences that need entity sentiment identification, along with a set of corresponding semantic mapping rules. By leveraging linguistic characteristics, we can effectively reduce error propagation and hallucination issues in the sentiment classification procedure of LLMs.

Specifically, as shown in Figure 2, let's take the example sentence $S$ := "The laptop's design is sleek, ..." and the aspect terms $T$ = "laptop's design" & "battery life" for presentation, we design and obtain LLM responses to the linguistic variants of the queries $Q$ := "From $S$, can we infer that the $T$ is positive/neutral/negative?".

```
# <PROMPT>
# Your task is to answer the following questions with "Yes" or "No" and input the answer into the slot ${PRED}.

# For "laptop's design":
Variant 1: "From 'The laptop's design is sleek and modern, ...', can we infer that the laptop's design is positive?" ${PRED}
Variant 2: "From 'The laptop's design is sleek and modern, ...', can we infer that the laptop's design is negative?" ${PRED}
Variant 3: "From 'The laptop's design is sleek and modern, ...', can we infer that the laptop's design is neutral?" ${PRED}

# For "battery life":
Variant 1: "From 'The laptop's design is sleek and modern, ...', can we infer that the battery life is positive?" ${PRED}
Variant 2: "From 'The laptop's design is sleek and modern, ...', can we infer that the battery life is negative?" ${PRED}
Variant 3: "From 'The laptop's design is sleek and modern, ...', can we infer that the battery life is neutral?" ${PRED}

# Let's solve these questions:
Copy and paste the initial problem here, making sure to replace the ${PRED} placeholder with the corresponding label, "Yes" or "No".
```

The above prompts displays the auto-generated prompts interacting with LLM of this example. We then use these prompts to interact with LLM to gather its feedback on each of these variant queries. The LLM provided responses in slots `${PRED}` indicates whether the sentiment hypothesis for each aspect term in each linguistic variants is correct or incorrect. Finally, we use these LLM-generated responses (i.e., the intermediate judgments) as emotion-oriented in-context constraints, serving as the context for the subsequent second stage.

Note that in the second stage, **In-contexts infusion & sentiment prediction**, to ensure the reliability of the LLM's responses, we employ a consistency alignment mechanism. This mechanism evaluates the intermediate responses for consistency. For instance, if the LLM confirms that "the laptop's design is positive/negative/neutral." if Yes/No/No → (positive); No/Yes/No → (negative); No/No/Yes → (neutral). Through this, we ensure these responses are logically consistent with the context of the original sentence. All these procedures are driven by Chain-of-Thought prompting, which directs the LLM through a systematic reasoning process. Additionally, we employ symbolic representations/premises, i.e., the first-order logic, to ensure the consistency and accuracy of the query mapping rules, reducing ambiguity and enhancing the reliability of the reasoning process.

### 4.2.2 | In-context Backgrounds Infusion Strategy

In-Context Learning (ICL) [10, 15, 16] is prominent prompting technique that performs exceptionally well in downstream NLU tasks. ICL has been the focus of significant study since its introduction. In the interaction with LLM, ICL is a prompt inference technique that, by simply concatenating input-label pairs from the training data and providing similar query cases in a list format, offers more valuable and accurate references for the LLM without requiring gradient updates. However, ICL presents several notable limitations [10, 16]. The LLM's prediction performance does not heavily rely on the ground truth input-label mappings within the demonstrations, leading to a marginal performance drop when labels are randomized. This suggests a limited dependence on the accuracy of input-label pairs, which can affect the robustness of ICL. Additionally, ICL's performance is significantly influenced by the distribution of the input text; using out-of-distribution (OOD) text in demonstrations causes a notable performance decline.

Similarly, in fine-grained classification tasks like ABSA, ICL faces specific challenges [15, 17]. The reliance on input distribution means that ABSA performance can suffer when the sentiment features are subtle or context-dependent. The structural importance over content accuracy in ICL may lead to misjudgments in sentiment classification, especially in complex or nuanced contexts. Therefore, it is essential to go beyond simple ICL techniques and design more robust, context-aware prompting strategies to ensure that LLMs accurately complete ABSA tasks.

To tackle these limitations, we introduce the In-context Backgrounds Infusion strategy for LLM prompting-based ABSA tasks. This strategy builds on the integration of external knowledge about the target entity or aspect term to significantly enhance the efficacy of emotion-oriented in-context techniques in sentiment analysis [21]. Inspired by the way humans incorporate background knowledge when evaluating sentiments, this strategy involves prompting LLMs to automatically generate brief introductions for each aspect term. These concise feature descriptions, termed in-context backgrounds, are then used as additional prompts to enhance the final sentiment determination. The detailed process of In-context Backgrounds Infusion strategy is as depicted follows.

Firstly, we design prompts that instruct the LLM to generate background information for the aspect terms. For example, given the example sentence $S$ := "The laptop's design is sleek, ..." and the aspect terms $T$ = "laptop's design" & "battery life" in Figure 2, we craft prompts like:

```
# <PROMPT>
# Given the sentence ${S} :=
"The laptop's design is sleek, but the battery life is barely lasts an hour."
and the aspect terms ${T} :=
"laptop's design" & "battery life",
# Please provide a brief introduction about the aspect terms, focusing on what makes it positive or
    negative for users.
# Following the response format for each aspect term:
% "${aspect term 1: introduction 1}",
% "${aspect term 2: introduction 2}",
...
```

These prompts are fed to the LLM, prompting it to generate background information relevant to each aspect term. These backgrounds help the LLM to better understand the aspect terms and the sentiments associated with them. For example, the LLM might generate responses like:

```
# <LLM RESPONSE>
% "laptop's design: Laptop design refers to the aesthetics and build quality of the device, often
    characterized by sleek lines, modern materials, and an overall appealing look that enhances user
    experience."
```

```
% "battery life: Battery life in laptops is crucial as it determines how long a laptop can be used without recharging. Poor battery performance can significantly hinder productivity and user satisfaction."
...
```

Finally, the generated background information is integrated into the LLM's context using the In-Context Learning (ICL) technique. This involves providing the LLM with these backgrounds as part of the input context for the original sentiment analysis task. By following this process, the In-context Backgrounds Infusion Strategy enhances the LLM's capability to perform nuanced sentiment analysis by leveraging relevant background information of aspect terms. This not only provides richer contextual information but also helps the LLM to understand the subtleties and specifics of each aspect term, leading to more accurate and context-aware sentiment judgments.

### 4.2.3 | In-context Analogies Infusion Strategy

Inspired by analogical reasoning in psychology, where humans utilize relevant past experiences to address new challenges [45], we aim to replicate this reasoning process in LLMs to enhance their problem-solving abilities in ABSA. When encountering a new sentiment analysis task, humans naturally ask themselves, "Have I encountered a similar situation before?" and draw upon how they navigated analogous situations to gain insights for the current problem. They also rely on high-level understanding, such as recognizing patterns of positive and negative feedback based on context.

In this part, specifically within the In-context Analogies Infusion Strategy, we prompt LLMs to autonomously generate relevant analogies that provide deeper context for sentiment analysis. Specifically, given a sentence to analyze, we instruct the LLM to recall similar scenarios and describe the sentiments involved, thus enriching the LLM's understanding. By integrating these analogies into the LLM's reasoning process, its ability to perform accurate and contextually aware sentiment analysis is significantly improved. The detailed process of In-context Analogies Infusion strategy is as depicted follows.

Firstly, we start by designing prompts that guide the LLM to recall and generate relevant analogies. For example, given the example sentence $S$ := "The laptop's design is sleek, ..." and the aspect terms $T$ = "laptop's design" & "battery life" in Figure 2, we craft prompts like:

```
# <PROMPT>
# Given the sentence ${S} :=
"The laptop's design is sleek, but the battery life is barely lasts an hour."
and the aspect terms ${T} :=
"laptop's design" & "battery life",
# Please recall relevant analogies, and provide similar cases like the above ${S} & ${T}.
# Describe the situation and the sentiments expressed about the design and battery life.
# Following the response format for each analogy:
%"${sentence};"
%% "${aspect term 1: sentiment 1, aspect term 2: sentiment 2, ...}".

%"${sentence};"
%% "${aspect term 1: sentiment 1, aspect term 2: sentiment 2, ...}".
...
```

These prompts are then fed to the LLM, prompting it to generate analogies relevant to the given aspect terms. For example, the LLM might generate responses like:

```
# <LLM RESPONSE>
% "A smartphone received high praise for its sleek and innovative design, but many users were disappointed with its short battery life, leading to mixed reviews."
%% "smartphone": positive, "battery life": negative.

% "A car was celebrated for its stylish design and comfortable interior, but it faced criticism due to its poor fuel efficiency."
%% "car": positive, "fuel": negative.
...
```

Finally, the generated analogies are integrated into the LLM's context using the In-Context Learning (ICL) technique, involving providing the LLM with these analogies as part of the input context for the original sentiment analysis task. By following this structured process, the In-context Analogies Infusion Strategy enhances the LLM's capability to perform nuanced sentiment analysis by leveraging relevant analogies and providing richer contextual information.

### 4.3 | In-Context Infusion & Sentiment Prediction

In the second stage, these obtained in-contexts are utilized as the upper context for In-Context Learning (ICL) to assist the LLM in making sentiment predictions.

Returning to the example in Figure 2, we integrate the emotion-oriented in-contexts obtained in the first stage with the original sentence $S$ and aspect terms $T$ using the Symbolic Chain-of-Thought and In-Context Learning techniques. These in-contexts serve as a comprehensive background for sentiment analysis of the original sentence. With this enriched context, the LLM can make more accurate and nuanced sentiment judgments.

Specifically, we achieve this through the following steps:

1. **Processing In-Context Constraints using Consistency Alignment Mechanism:** To ensure the consistency of the reasoning process, we use the Symbolic Chain-of-Thought (SymbCoT) module. This module employs first-order logic to convert multiple variant constraints into consistent sentiment judgments. By leveraging the SymbCoT module, which uses first-order logic, we enhance the reasoning and prediction accuracy. This ensures logical consistency in the reasoning process, reduces ambiguity, and improves the reliability of judgments.

For example, if the LLM confirms that the "laptop's design is positive/negative/neutral," then: "yes/no/no" → (positive); "no/yes/no" → (negative); "no/no/yes" → (neutral). This can be formally represented with the following conversion rules (logical expressions):

## Positive Sentiment Logical Expression: $(P_{\text{pos}} \wedge \neg P_{\text{neg}} \wedge \neg P_{\text{neu}})$, i.e., $P_{pos}\&\&!P_{neg}\&\&!P_{neu}$.

## Negative Sentiment Logical Expression: $(P_{\text{neg}} \wedge \neg P_{\text{pos}} \wedge \neg P_{\text{neu}})$, i.e., $!P_{pos}\&\&P_{neg}\&\&!P_{neu}$.

## Neutral Sentiment Logical Expression: $(P_{\text{neu}} \wedge \neg P_{\text{pos}} \wedge \neg P_{\text{neg}})$, i.e., $!P_{pos}\&\&!P_{neg}\&\&P_{neu}$.

These logical expressions ensure that the sentiment judgments are consistent and accurate by verifying the truth values of the constraints and converting them into a unified sentiment prediction.

2. **Integration of Emotion-Oriented In-Contexts:** Firstly, the emotion-oriented in-contexts obtained from the first stage, including processed in-context constraints from consistency alignment mechanism, in-context backgrounds, and in-context analogies, are integrated as the background for the initial query.

For instance, for the sentence $S$ := "The laptop's design is sleek and modern, but the battery life is disappointing and barely lasts an hour." and aspect terms $T$ := "laptop's design" and "battery life", the following is an example of how such a prompt looks:

```
# ${ASPECT} := "laptop's design",
# ${SENT} := "The laptop's design is sleek and modern, but, ...".

# ${Constraints}:
# For "laptop's design":
- From "The laptop's design is sleek and modern, ...", can we infer that the laptop's design is positive? Yes
- From "The laptop's design is sleek and modern, ...", can we infer that the laptop's design is negative? No
- From "The laptop's design is sleek and modern, ...", can we infer that the laptop's design is neutral? No

# Based on this,
the first constraint is Yes, thus Ppos := true;
the second constraint is No, thus Pneg := false;
the third constraint is No, thus Pneu := false.
# Then,
Positive: Ppos && !Pneg && !Pneu := true;
Negative: !Ppos && Pneg && !Pneu := false;
Netural: !Ppos && !Pneg && Pneu := false.

Finally, the sentiment for the laptop's design is likely positive.

# ${Backgrounds}:
# - "laptop's design": The design of the laptop is sleek and modern, characterized by its clean lines and contemporary aesthetics.

# ${Analogies}:
# - "laptop's design": A smartphone with a sleek design received positive feedback for its appearance, similar to how the laptop's design is appreciated.

# ${Query}:
```

```
# - Given the above contexts, ${Constraints}, ${Backgrounds}, and ${Analogies}, your task is to
    identify the sentiment polarity of ${ASPECT} in ${SENT}. positive/ negative/ neutral?
```

By integrating these constraints, backgrounds, and analogical reasoning, we provide the LLM with rich and consistent contexts for predicting sentiment. By following this above process, we ensure that the LLM has a comprehensive understanding of the context, leading to more accurate and reliable sentiment predictions.

## 4.4 | Self-Adaptive Bootstrap Instructions Optimization

To enhance the performance and robustness of our BYD-OBS-ABSA framework, we employ a Self-Adaptive Bootstrap Instructions Optimization method. The primary goal is to systematically fine-tune the prompts and demonstrations provided to large language models (LLMs), ensuring they generate more accurate and reliable sentiment predictions. This is crucial because optimizing the way LLMs are instructed can significantly impact their ability to understand and process nuanced sentiment information in ABSA tasks.

**Algorithm 1** Self-Adaptive Bootstrap Instructions Optimization

**Input:** DSPy program $P$, training set $T$, validation set $V$, performance metric $M$, number of trials $N$

**Output:** Optimized program $P^*$

Initialize configuration parameters: max bootstrapped demos, max labeled demos, num candidate programs, etc.
Initialize an empty list of candidate programs and their scores
**for** $i = 1$ to $N$ **do**
  Shuffle the training set $T$
  Initialize a temporary program $P_i$ as a copy of $P$
  **for** each stage in $P_i$ **do**
    Extract a subset of labeled examples from the shuffled training set
    Generate bootstrapped demonstrations by:
    - Using the current model to predict outputs for the extracted examples
    - Validating the predicted outputs against the performance metric $M$
    - Selecting the top predictions that meet the metric criteria
    Combine the bootstrapped demonstrations with labeled examples to form the few-shot examples for this stage
    Update the instructions and examples for the current stage with the newly formed few-shot examples
  **end for**
  Evaluate $P_i$ on validation set $V$ using metric $M$
  Record the score of $P_i$ and add it to the list of candidate programs
**end for**
Select the program $P^*$ with the highest score from the list of candidate programs
**return** Optimized program $P^*$

Self-Adaptive Bootstrap Instructions Optimization is a bootstrap technique with random search for few-shot optimization, combining the principles of few-shot learning and random search to iteratively refine the instructions and examples provided to the LLM. As shown in Algotithm 1, the Self-Adaptive Bootstrap Instructions Optimization operates through the following steps:

We initialize configuration parameters essential for bootstrapping, including the number of bootstrapped demonstrations and labeled examples, as well as the number of candidate programs to be generated and evaluated. In each trial, the training set is shuffled to create a diverse set of examples, and a temporary program is initialized as a copy of the original program. For each stage of the temporary program, a subset of labeled examples is extracted from the shuffled training set. Bootstrapped demonstrations are generated by using the current model to predict outputs for these examples, validating these outputs against the performance metric, and selecting the top predictions that meet the criteria. These demonstrations are then combined with labeled examples to form the few-shot examples for the current stage, and the instructions and examples are updated accordingly. Each candidate program is evaluated using the specified performance metric, and scores are recorded. After all trials, the candidate program with the highest score is selected as the optimized program, ensuring the best configuration of instructions and examples. The optimized program is then returned, ready for further tasks or deployment in production environments.

This optimization strategy ensures that the language model is provided with the most effective and contextually relevant instructions, leading to more accurate and reliable sentiment predictions in the BYD-OBS-ABSA framework.

Moreover, to standardize and optimize the design of prompts and information processing workflows within BYD-OBS-ABSA, we utilize Stanford University's open-source LM prompts optimizing toolkit, DSPy[2]. DSPy [46, 47] is a programming model that

**TABLE 1** The data split statistics of these adopted ABSA benchmarks.

| Dataset | Train | | | Dev | Test | Total |
|---|---|---|---|---|---|---|
| | Initial/full-data | Zero-shot | Few-shot | | | |
| rest14 | 1796 | 0 | 1/3/6/9 | 181 | 601 | 2578 |
| lap14 | 1308 | 0 | 1/3/6/9 | 150 | 411 | 1869 |
| rest15 | 750 | 0 | 1/3/6/9 | 82 | 400 | 1232 |
| rest16 | 1113 | 0 | 1/3/6/9 | 118 | 420 | 1651 |
| Twitter | 1687 | 0 | 1/3/6/9 | 422 | 233 | 2342 |
| ACLShortData | 5623 | 0 | 1/3/6/9 | 625 | 692 | 6940 |

abstracts LM pipelines as text transformation graphs. Its most notable feature is its parameterization, which enables it to learn how to apply a combination of prompting, fine-tuning, augmentation, and reasoning techniques for LLMs by creating and collecting demonstrations.

## 5 | EXPERIMENTS

### 5.1 | Datasets

For our experimental evaluation, we use six real-world benchmark datasets, each meticulously annotated with aspect categories, aspect terms, and sentiment polarities, making them ideal for aspect-based sentiment analysis (ABSA). Below is an overview of these benchmarks:

- **SemEval-2014** [32]: Provided by the SemEval-2014 competition[3], this dataset includes benchmarks for both the restaurant and laptop domains, widely used in ABSA research.
- **SemEval-2015** [48]: As an extension of the SemEval-2014 restaurant dataset, the SemEval-2015 dataset was released by the workshop organizers[4].
- **SemEval-2016** [49]: This dataset, continuing from previous SemEval competitions[5], focuses on the restaurant domain and is proposed by Pontiki et al. (2016).
- **Twitter** [50]: Collected and annotated by Mitchell et al.[6] and later reorganized by Li et al. [**?** ], this dataset includes tweets from the Twitter platform.
- **ACLShortData** [51]: Introduced at ACL 2014[7], this dataset is based on social tweets annotated via Amazon Mechanical Turk.

These datasets from the SemEval competitions [32, 48, 49] are among the most frequently cited domain-specific benchmarks for ABSA, capturing user evaluations of products and services in the laptop and restaurant domains. It is noteworthy that while the SemEval-2015 and SemEval-2016 datasets are smaller than the SemEval-2014 dataset, the SemEval-2016 dataset is particularly diverse, covering five domains and eight languages, though we use only the English restaurant domain for our experiments. The ACLShortData comprises around 10,000 English tweets and 30,000 Spanish tweets; however, our experiments are based solely on the English portion.

Table 1 summarizes the statistical details of these datasets. Here, '*rest14*' and '*lap14*' refer to the restaurant and laptop subsets from the SemEval-2014 competition (task 4), while '*rest15*' and '*rest16*' pertain to the datasets from the SemEval-2015 (task 12) and SemEval-2016 (task 5) competitions, respectively. For each dataset, the zero-shot column indicates that no training examples are provided, whereas the few-shot column shows the number of training examples used in few-shot learning scenarios (where K = 5, 20, or 50 examples). This partitioning helps to evaluate the model's performance with minimal training data.

### 5.2 | Baselines

In our experiments, we evaluate several models designed for ABSA, including both traditional small language models (SLMs) and recent large language models (LLMs).

**BERT-SPC** [24] refers to the Sentence Pair Classification task of the pre-trained LM, BERT[8]. In BERT-SPC, the input sequence is constructed as "[CLS]" + global context + "[SEP]" + [asp] + "[SEP]" and then fed into the basic BERT model for sentence pair

classification task;

**BERT-BiLSTM-CRF** [52] combines BERT's contextual semantic understanding, BiLSTM's sequence feature extraction, and CRF's sequence labeling capabilities, making it well-suited for tasks like named entity recognition. Such a integration allows the model to accurately capture and predict complex language patterns in text;

**LCF-ABSA** [23] represents a significant Local Context Focus Aspect-Based Sentiment Analysis model by effectively combining local and global context features. Semantic-Relative Distance (SRD) is introduced to evaluate the dependency between contextual words and aspects. The Local Context Focus design incorporates a pretrained BERT model. And Two specific layers, Context-Dependent Masking (CDM) and Context-Dependent Weighting (CDW) Layers, is implemented to enforce the local context focus;

**BERT-PT** [27] enhances the performance of BERT on domain-specific tasks like Review Reading Comprehension (RRC) and aspect-based sentiment analysis by leveraging a novel post-training technique. This approach adapts BERT with both domain knowledge from unsupervised domain reviews and task knowledge from supervised MRC data;

**AEN-GloVe & AEN-BERT** [24] introduce label smoothing regularization to handle label unreliability and incorporates GloVe word embeddings and pre-trained BERT to enhance performance. AEN-GloVe/-BERT addresses the limitations of traditional RNN-based approaches by using attention-based encoders to model the relationship between context and target words without relying on recurrence.;

**BERT-ADA** [22] is a deep transfer-learning-based ABSA method, namely Domain Adaptation through BERT LM, that employs a two-step procedure: self-supervised domain-specific BERT fine-tuning followed by supervised task-specific fine-tuning;

**Direct Ask on GPT-3.5-turbo** (*abbr.*, vanilla GPT) This approach involves using `GPT-3.5-turbo` [10, 12] directly for ABSA tasks without additional prompting techniques;

**Chain-of-Thought on GPT-3.5-turbo** (*abbr.*, CoT-GPT) Employs Chain-of-Thought prompting [13, 14] on `GPT-3.5-turbo`, which involves guiding the LLM through a systematic reasoning process by generating intermediate reasoning steps to improve accuracy and explainability;

**In-context Learning on GPT-3.5-turbo** (*abbr.*, ICL-GPT) This utilizes In-Context Learning prompting [15] on top of `GPT-3.5-turbo` where the LLM is provided with example input-label pairs during inference to offer valuable references for similar tasks.

**Direct Prediction using GPT-4.0** (*abbr.*, vanilla GPT-4) This utilizes the superior GPT-4.0 [11] for direct prediction in ABSA. GPT-4.0 offers significant improvements over GPT-3.5, including enhanced contextual understanding and more accurate handling of complex queries. These advancements allow GPT-4.0 to provide more reliable and precise sentiment predictions.

## 5.3 | Evaluation Metric

To maintain consistency with previous studies, this research utilizes accuracy as the primary evaluation metric, as it is the most commonly used performance metric in aspect-level sentiment analysis (ASBA) research, rather than precision, recall, or F1 score [1, 2]. In general, a well-designed ABSA method should exhibit improved performance, which is reflected in higher accuracy scores. Accuracy serves as a standard measure for tasks that can be framed as label classification problems. It indicates the proportion of correctly predicted instances out of the total number of instances and is defined by the following equation:

$$\text{Accuracy} = \frac{Count_T}{Count_N} \tag{1}$$

Here, $Count_T$ represents the number of correctly predicted instances, while $Count_N$ signifies the total number of instances. For each instance evaluated, a prediction is considered correct if the predicted sentiment polarity matches the actual sentiment polarity. If the predicted sentiment does not match, it is considered incorrect.

## 5.4 | Experimental Settings

To ensure the transparency and reproducibility of our experiments, we utilized Google Colab[9], an online cloud-based platform for running and sharing code. Colab provides a convenient environment for executing experiments and sharing results.

The virtual machine configuration used for our experiments is as follows: Operating System: Ubuntu 22.04.3 LTS; CPU: Intel(R) Xeon(R) CPU @ 2.00GHz; GPU: Tesla T4 with 16GB memory; CUDA Version: 12.2. Our approach primarily focuses on designing better LLM prompting engineering. Except for the *Self-Adaptive Bootstrap Instructions Optimization* phase, which requires CPU participation for bootstrap optimization, other LLM prompting processes can be categorized under direct network API calls. Therefore, the experimental hardware requirements are not stringent.

In our experiments, we set various parameters to optimize the performance of BYD-OBS-ABSA by using the *Self-Adaptive Bootstrap Instructions Optimization*. We configured the maximum number of bootstrapped demonstrations to 4 and the maximum number of labeled demonstrations to 16. The number of training and evaluation rounds was limited to 1. We set the number of candidate programs to 16 and utilized 6 threads for parallel processing. To manage errors, we allowed a maximum of 10 errors. We did not specify a particular score threshold for early stopping, nor did we set a specific metric threshold for performance evaluation.

For the selection of few-shot examples, to ensure the fairness and persuasiveness of the experiment, as well as to maintain the balance of sample labels, we fixed the selection of the first $K$ training samples from each category label in the train set of each benchmark for $K$-shot settings ($K \in \{1/3/6/9\}$). It is noteworthy that the experimental results are influenced by the current performance and version of the officially released LLMs, as well as the stochastic nature of the LLM's responses. As a result, the outcomes may exhibit some variability. However, this variability is expected to be within a reasonably acceptable range. Our GitHub repository[10] contains not only the code implementation of the BYD-OBS-ABSA operational framework, but also the LLM querying processing records for each example tested on all benchmark test sets, along with the comparison between the final predicted labels and the gold labels. Moreover, although there may be slight variations in the definition and settings of prompts during the experiment, the overall prompting schema remains consistent. For detailed construction processes, please refer to the prompts in the *Colab Jupyter Notebooks* on our released codes.

## 5.5 | Main Results

In this study, we primarily compare the ABSA performance results of our BYD-OBS-ABSA with those of various state-of-the-art (SOTA) baselines on several widely used datasets in both zero-shot and few-shot settings. This comparison allows us to highlight the strengths and weaknesses of BYD-OBS-ABSA, providing a comprehensive understanding of its characteristics.

## 5.6 | Zero-Shot Performance

Table 2 presents a comparative summary of the proposed BYD-OBS-ABSA performance against the baseline methods. In this table, for the implementation of the GPT-3.5-based in-context learning method (ICL-GPT), we selected three examples from the training set that correspond to the first occurrences of three distinct emotional categories (each in-context instance corresponds to a single sentiment category) to serve as the context for this prompting approach. Moreover, all models under the SLM-based fine-tuning paradigm are trained on full data, aligning with the same data distribution. Therefore, their predictive performance is undoubtedly superior and not comparable to LLM prompting in a zero-shot setting. Consequently, the following analysis will focus more on the comparisons and conclusions drawn from different models based on the LLM prompting paradigm.

From the overall zero-shot results in Table 2, our proposed BYD-OBS-ABSA framework shows a significant advantage over other baselines, such as CoT-GPT and ICL-GPT on most benchmarks. This underscores the efficacy of our core idea of *Emotion-Oriented In-Contexts Integration* in BYD-OBS-ABSA for zero-shot learning ABSA scenarios.

At first glance, BYD-OBS-ABSA excelled on the Rest15 dataset with an accuracy of 82.2%, significantly higher than CoT-GPT's 57.00%, marking a 25.2% improvement. On the Rest16 dataset, BYD-OBS-ABSA achieved an accuracy of 91.50%, surpassing CoT-GPT's 90.00%, indicating a 1.5% improvement. BYD-OBS-ABSA also performed well on the Twitter dataset with an accuracy of 54.90%, surpassing CoT-GPT's 44.64% by 10.26%. BYD-OBS-ABSA achieved an accuracy of 53.54% on the ACLShortData, which is much higher than CoT-GPT's 26.16%, reflecting a 27.38% improvement.

On the Rest14 and Lap14 datasets, BYD-OBS-ABSA showed improvements of 0.64% and 1.38% over CoT-GPT, respectively. While these improvements are positive, they are not very significant. We think this can be attributed to: 1. Both the Rest14 and Lap14 datasets might contain reviews with relatively straightforward sentiment expressions that CoT-GPT can handle effectively, making the added value of emotion-oriented contexts in BYD-OBS-ABSA less pronounced. 2. The Rest14 and Lap14 datasets may contain specific distribution patterns and domain-specific terminologies that are better captured with models having access to training data. 3. Moreover, another plausible explanation is that the extensive original corpora used in the pre-training process of the GPT-3.5-turbo model might encompass data similar to the Rest14 and Lap14 datasets. This could potentially lead to effective sentiment prediction using simple prompting methods such as Direct Ask or CoT, while more sophisticated approaches like BYD-OBS-ABSA may introduce additional intrinsic errors.

On Rest16, BYD-OBS-ABSA achieved an accuracy of 91.50%, surpassing CoT-GPT's 90.00%, indicating a 1.5% improvement. Additionally, BYD-OBS-ABSA outperformed Direct Ask on GPT-3.5 and ICL-GPT, which had accuracies of 90.24% and 87.62%, respectively. This demonstrates the robust performance of BYD-OBS-ABSA, even in comparison to other sophisticated models. These results highlight the effectiveness of our BYD-OBS-ABSA. The Rest16 benchmark heavily depend on the training data distribution, making it challenging for zero-shot approaches. However, BYD-OBS-ABSA's performance suggests it effectively incorporates diverse contextual insights, allowing it to handle the specific distribution patterns and domain-specific terminologies present

in Rest16. The superior accuracy of BYD-OBS-ABSA, even in zero-shot settings, underscores its potential for sentiment prediction tasks across various benchmarks.

To summarize, BYD-OBS-ABSA exceeds most advanced LLM-based methods in zero-shot settings across diverse datasets. Its capacity to integrate comprehensive contextual insights and refine prompting instructions positions it as a robust and superior choice for ABSA tasks.

To further improve performance on datasets like Rest14 and Lap14, future strategies could involve incorporating domain-specific pre-trained embeddings and fine-tuning on a smaller subset of domain-specific data. Additionally, enhancing the model's ability to recognize and adapt to specific distribution patterns present in the dataset could further boost its performance. In the future, we also plan to conduct Parameter-Efficient Fine-Tuning (PEFT) on LLMs such as GPT-3.5 to further improve their ability to adapt to specific datasets.

**TABLE 2** The performance statistics (Metric: % Accuracy) between our BYD-OBS-ABSA framework and the state-of-the-art (SOTA) baselines. The highest performance is highlighted in bold, while the second-best results are underlined. Results marked with an asterisk (*) were obtained from previous works.

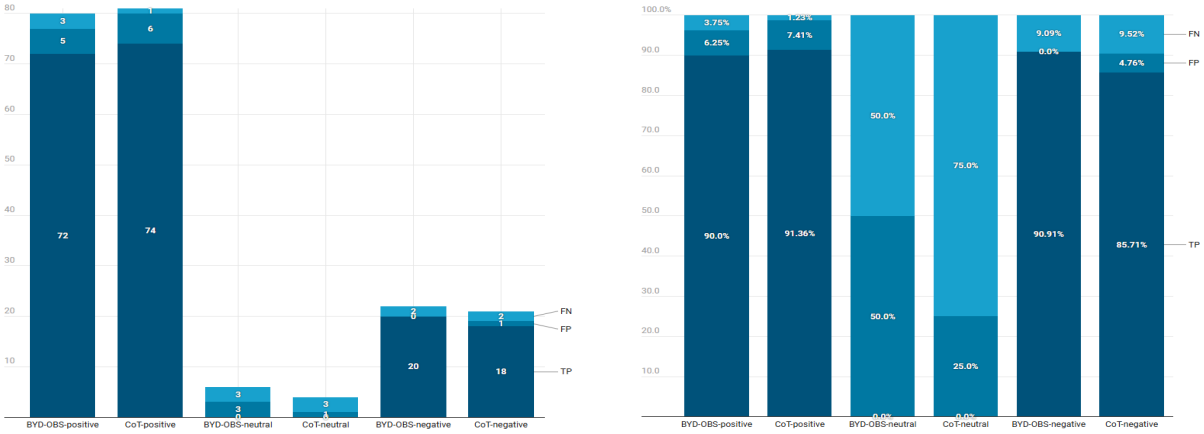
| **Methods / benchmarks** | | Rest14 | Lap14 | Rest15 | Rest16 | Twitter | AclShortData |
|---|---|---|---|---|---|---|---|
| | | Acc (%) | Acc (%) | Acc (%) | Acc (%) | Acc (%) | Acc (%) |
| **SLM-based Fine-tuning** (trained on full data) | | | | | | | |
| BERT-SPC | | 81.0 | 79.0 | 88.7 | 87.3 | 76.0 | 73.5 |
| BERT-BiLSTM-CRF | | 87.3 | 79.9 | **90.3** | 88.5 | 76.4 | 70.9 |
| WordTransABSA* | | 85.7 | 78.4 | 89.1 | 89.0 | 76.1 | 72.5 |
| BERT-PT* | | 85.0 | 78.1 | 90.0 | 87.7 | - | - |
| AEN-GloVe* | | 81.0 | 73.5 | - | - | - | 72.8 |
| AEN-BERT* | | 83.1 | 79.9 | - | - | - | 74.7 |
| BERT-ADA* | | **87.7** | 79.9 | **90.3** | 88.2 | 75.8 | 71.6 |
| LCF-ABSA* | | 87.1 | **82.5** | 90.1 | 88.9 | 76.6 | 72.7 |
| **LLM-based Prompting** (Zero-Shot Reasoning) | | | | | | | |
| | All Samples (Test) | 600 | 411 | 400 | 420 | 233 | 691 |
| Direct Ask on GPT-3.5 | Acc (%) | 79.20 | 75.71 | 79.00 | 90.24 | 48.07 | 49.57 |
| -turbo (vanilla GPT) | Corrected Samples | 316 | 159 | 316 | 379 | 112 | 343 |
| Chain-of-Thought on | Acc (%) | 81.50 | 77.64 | 57.00 | 90.00 | 44.64 | 26.16 |
| GPT-3.5-turbo (CoT-GPT) | Corrected Samples | 489 | 315 | 228 | 378 | 104 | 181 |
| In-context Learning on | Acc (%) | 81.50 | 77.13 | 78.75 | 87.62 | 49.79 | 50.14 |
| GPT-3.5-turbo (ICL-GPT) | Corrected Samples | 489 | 317 | 315 | 368 | 116 | 347 |
| **Our Model** (Zero-Shot Reasoning) | | | | | | | |
| BYD-OBS-ABSA | Acc (%) | **82.14** | **79.02** | **82.20** | **91.50** | **54.90** | **53.54** |
| (GPT-3.5-turbo) | Corrected Samples | **492** | **324** | **329** | **384** | **128** | **370** |

To delve deeper into the performance characteristics of BYD-OBS-ABSA, we selected the Rest14, Rest16, and Twitter datasets for an extensive predictive feature analysis. This analysis aimed to ascertain the distinct advantages and disadvantages exhibited by our BYD-OBS-ABSA, compared to other baselines like CoT-GPT. We meticulously calculated the distribution and percentage of True Positives (TP), False Positives (FP), and False Negatives (FN) across three sentiment categories, as summarized in Figure 3. Note that:

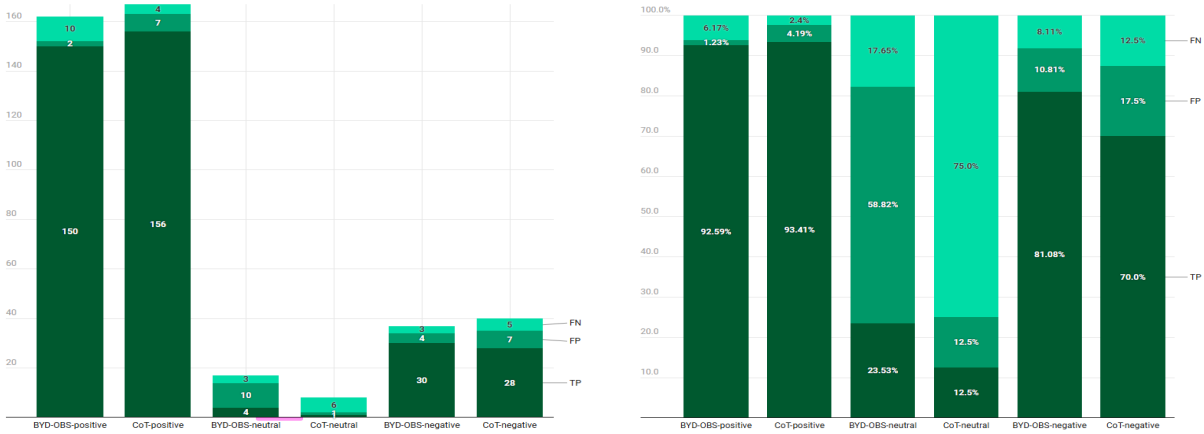
1. **High FP Rate**: A high FP rate might indicate that the model occasionally misclassifies neutral or negative sentiments as positive. This could be due to overly generalized training data or inherent biases in the language model.

2. **High FN Rate**: A high FN rate, especially in the negative category, might suggest the model's difficulty in detecting subtler expressions of dissatisfaction or critique. This is critical for nuanced sentiment analysis.

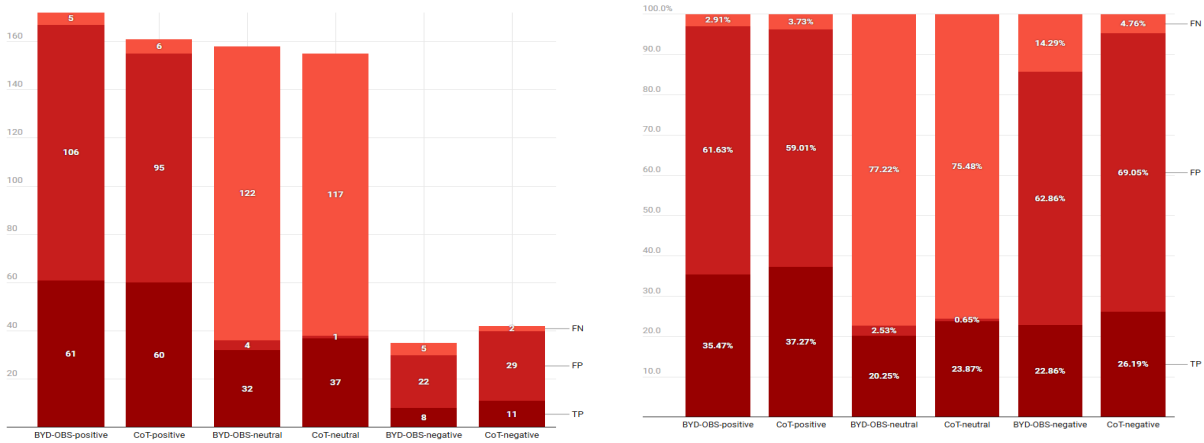
The subfigures (a), (b), and (c) within Figure 3 correspond to the evaluation results from each respective dataset. Each subfigure is divided into two parts: The left part illustrates the count distribution of TP, FP, and FN ratios across the positive, neutral, and negative sentiment categories. In this visualization, the horizontal axis categorizes the sentiment categories as delineated by different models, while the vertical axis enumerates the counts of each prediction feature (i.e., TP, FP, and FN) associated with these sentiment categories. The right part maintains the horizontal alignment with the left, detailing the percentage share of predictions for each sentiment category relative to the prediction features (i.e., TP, FP, and FN).




a). Sentiment Prediction Distribution Comparison on the **Rest14** Benchmark




b). Sentiment Prediction Distribution Comparison on the **Rest16** Benchmark




c). Sentiment Prediction Distribution Comparison on the **Twitter** Benchmark

**FIGURE 3** Comparative analysis of sentiment prediction feature distribution: a statistical examination of our byd-obs-absa and the cot-gpt baseline on **Rest14**, **Rest16**, and **Twitter** benchmarks. Across positive, neutral, and negative categories with respect to TP, FP, and FN ratios. Left: Visualization of different predicted distribution quantities; Right: Visualization of predicted distribution percentages.

By examining the distribution and percentages of TP, FP, and FN across sentiment categories (positive, neutral, negative), we can deduce several aspects of the model's performance:

- Firstly, a high TP rate in specific categories indicates that this model effectively identifies and correctly classifies sentiments in those categories. For exmaple, as illustrated by the right of subfigure (a) of the **Rest14** benchmark, BYD-OBS-ABSA achieves a remarkable TP ratio of 90.91% for negative sentiments, significantly outperforming CoT-GPT, which has a TP ratio of 85.71%. This suggests that BYD-OBS-ABSA is particularly adept at identifying negative sentiments in restaurant reviews.
- BYD-OBS-ABSA also demonstrates excellent performance in recognizing neutral and negative sentiments on the **Rest16** benchmark. It achieves TP ratios of 23.53% and 81.08% for neutral and negative sentiments, respectively, surpassing CoT-GPT's 12.5% for neutral and 70.00% for negative sentiments. This indicates a strong ability to detect a range of sentiments accurately.
- From subfigure (b), BYD-OBS-ABSA consistently shows lower FN rates than CoT-GPT across the neutral and negative categories, it suggests robustness in minimizing missed classifications, enhancing reliability in scenarios requiring high accuracy in sentiment detection.

The above phenomenons not only underscore the technical strengths of BYD-OBS-ABSA but also contextualizes its practical applications and effectiveness in real-world sentiment analysis tasks across different platforms and data sets. Incorporating emotion-oriented in-contexts significantly enhances the interpretative and reasoning capabilities of BYD-OBS-ABSA, leading to improved performance and accuracy in aspect-based sentiment analysis.

## 5.7 | Zero-shot Results Compared with Advaned GPT-4.0

We further investigated whether our BYD-OBS-ABSA framework, utilizing the relatively smaller and less powerful GPT-3.5 (approximately 175 billion parameters), could surpass the state-of-the-art GPT-4.0 (approximately 1.7 trillion parameters). GPT-4.0, with a parameter size about ten times that of GPT-3.5 and a much larger pre-training corpus, demonstrates significantly stronger language comprehension and generation capabilities. Our aim is to verify if BYD-OBS-ABSA can outperform GPT-4.0 in sentiment analysis through optimized prompting techniques, thereby showcasing the advantages of our prompting strategies.

Based on the zero-shot results comparison depicted in Figure 4, we analyzed the detailed performance of BYD-OBS-ABSA and GPT-4.0 in sentiment analysis on the **ACLShortData** and **Twitter** benchmarks.

In **ACLShortData**, with a total sample size of 691, BYD-OBS-ABSA correctly predicted 370 instances and incorrectly predicted 321 instances, indicating a generally good performance. In contrast, the direct use of GPT-4.0 resulted in 332 correct predictions and 359 incorrect predictions, highlighting strengths and weaknesses in different aspects of the task for both methods. From the percentage chart, BYD-OBS-ABSA achieved an accuracy of 54% and an error rate of 46%, while GPT-4.0 had an accuracy of 48% and an error rate of 52%. This further illustrates the varying strengths and weaknesses of the two methods across different evaluation metrics.

For **Twitter**, with a total sample size of 233, BYD-OBS-ABSA correctly predicted 128 instances and incorrectly predicted 105 instances. In comparison, the direct use of GPT-4.0 resulted in 106 correct predictions and 127 incorrect predictions. This indicates that while the overall performance of both methods is similar, there are differences in the numbers of correct and incorrect predictions. In the percentage chart, BYD-OBS-ABSA achieved an accuracy of 55% and an error rate of 45%, whereas GPT-4.0 had an accuracy of 45% and an error rate of 55%. This shows that BYD-OBS-ABSA has an advantage in accuracy on the Twitter dataset, while GPT-4.0 performs better in terms of error rate.

Overall, BYD-OBS-ABSA significantly outperforms direct prediction using GPT-4.0 on these two datasets. The BYD-OBS-ABSA framework enhances sentiment analysis accuracy and robustness by incorporating emotion-oriented contexts, unique contextual constraints, backgrounds, and analogical reasoning to optimize LLM predictions. In contrast, GPT-4.0 relies on its powerful pre-trained model, excelling in some tasks but potentially requiring further contextual optimization and prompting design for complex sentiment analysis tasks.

This indicate that even with the smaller and less powerful GPT-3.5 model, the BYD-OBS-ABSA framework can surpass GPT-4.0 in many ABSA tasks through carefully designed prompting techniques. This underscores the advantage of our prompting strategies in handling sentiment recognition tasks and demonstrates that optimizing prompt design can significantly enhance model performance, even for smaller models. Future research can further combine the strengths of both methods to explore more efficient sentiment analysis frameworks, achieving better performance across various ABSA datasets.

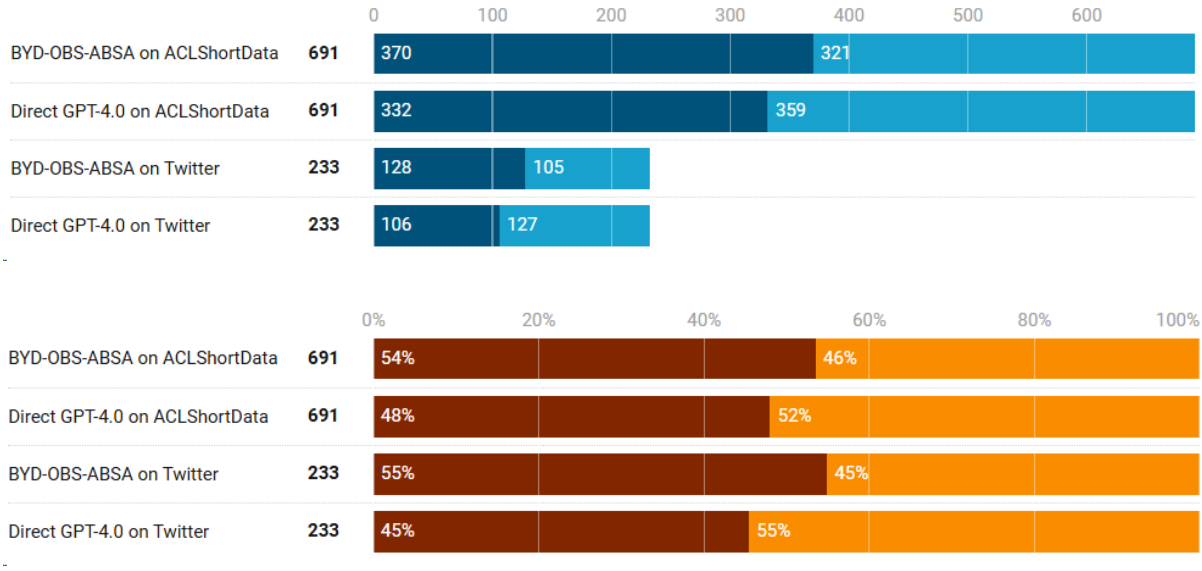



**FIGURE 4** The prediction results of BYD-OBS-ABSA and GPT-4.0 on ACLShortData and Twitter datasets. Top: displaying the correct and incorrect predictions (total examples shown on the left); Bottom: illustrating the accuracy rates (in percentage). BYD-OBS-ABSA generally outperforms Directly Asking using GPT-4.0 across both datasets.

## 5.8 | Few-Shot Performance

To evaluate the performance variation of our BYD-OBS-ABSA framework under limited labeled data scenarios, we conducted few-shot learning experiments. By comparing the actual performance of our BYD-OBS-ABSA against two other baseline models — Vanilla GPT (Direct Ask on GPT-3.5) and CoT-GPT (Chain-of-Thought on GPT-3.5-turbo) — across six benchmarks (rest14, lap14, rest15, rest16, Twitter, AclShortData) under few-shot settings of K=1, 3, 6, and 9, we aimed to understand how well each model leverages minimal training data.

It is worth noting that: The few-shot learning approach was conducted by choosing the top K samples for each sentiment category from the sorted datasets. This top K samples selection strategy for each sentiment category from the sorted dataset ensures a balanced and representative sample set. This systematic approach enhances the model's ability to generalize from a few examples and ensures reproducibility and fairness in the experiments.

Figure 3 presents the accuracy (%) of the three models across the datasets under the different few-shot settings. The overall results in Figure 3 indicate a clear and consistent trend: BYD-OBS-ABSA outperforms both Vanilla GPT and CoT-GPT across all datasets and K-shot settings. These demonstrate the superior performance of BYD-OBS-ABSA, which consistently achieves higher accuracy compared to the other models, particularly as the number of shots increases.

In the K=1 setting (Row Group 1 of Figure 3), BYD-OBS-ABSA exhibits significant improvements over the other models. For instance, on the Twitter dataset, it achieves an accuracy of 59.23%, compared to 50.21% for Vanilla GPT and 52.36% for CoT-GPT. As the number of shots increases to K=3 (Row Group 2), BYD-OBS-ABSA continues to show strong performance, achieving 87.17% accuracy on the rest14 dataset, compared to 82.83% for Vanilla GPT and 85.33% for CoT-GPT.

This trend persists in the K=6 setting (Row Group 3 of Figure 3), where BYD-OBS-ABSA demonstrates robust performance improvements across all datasets. Notably, it scores 86.50% on the rest15 dataset, a marked improvement over the 84.50% and 83.50% achieved by Vanilla GPT and CoT-GPT, respectively. At the highest K-shot setting (K=9, Row Group 4), BYD-OBS-ABSA maintains its superior performance, achieving the highest accuracy of 89.50% on the rest14 dataset and a significant lead on Twitter with 78.54%.

The consistent outperformance of BYD-OBS-ABSA can be attributed to several key mechanisms integrated within the framework. Firstly, the use of emotion-oriented in-contexts—including constraints, backgrounds, and analogies—provides a richer and more comprehensive context for sentiment analysis. This approach effectively mitigates the issues of LLM hallucinations and error propagation, which are common in other methods. Secondly, the self-adaptive bootstrap optimization enhances the model's ability to refine its instructions and examples iteratively, leading to more accurate predictions. This iterative refinement is particularly beneficial in few-shot settings where the amount of training data is limited, allowing the model to adapt and optimize based on the few available examples.

In contrast, Vanilla GPT relies solely on the model's pre-trained knowledge without additional prompting techniques, leading to lower performance in few-shot settings. It lacks the structured reasoning process provided by CoT-GPT and the rich contextual

**TABLE 3** Few-Shot Performance Comparison under different few-shot settings (K=1, 3, 6, 9) of BYD-OBS-ABSA with Other Baselines, i.e., vanilla GPT and CoT-GPT. The Evaluation Metric is also Accuracy (%).

| Few-Shot Settings (K-shot) | Models | rest14 | lap14 | rest15 | rest16 | Twitter | AclShortData |
|---|---|---|---|---|---|---|---|
| K=1 | Vanilla GPT | 81.50 | 77.37 | 79.00 | 87.86 | 50.21 | 50.22 |
| | CoT-GPT | 83.83 | 79.81 | 70.50 | 87.14 | 52.36 | 37.92 |
| | BYD-OBS-ABSA | **86.83** | **83.94** | **80.50** | **92.62** | **59.23** | **64.54** |
| K=3 | Vanilla GPT | 82.83 | 82.48 | 83.50 | 89.52 | 59.23 | 62.66 |
| | CoT-GPT | 85.33 | 82.24 | 79.50 | 90.24 | 61.80 | 46.02 |
| | BYD-OBS-ABSA | **87.17** | **85.40** | **83.00** | **92.62** | **63.95** | **70.48** |
| K=6 | Vanilla GPT | 85.33 | 83.45 | 84.50 | 89.52 | 66.95 | 69.46 |
| | CoT-GPT | 84.33 | 82.48 | 83.50 | 90.24 | 62.23 | 51.23 |
| | BYD-OBS-ABSA | **87.33** | **85.40** | **86.50** | **92.86** | **72.53** | **76.27** |
| K=9 | Vanilla GPT | 87.33 | 84.91 | 83.75 | 90.91 | 68.24 | 71.35 |
| | CoT-GPT | 88.17 | 84.43 | 84.50 | 91.67 | 72.96 | 59.62 |
| | BYD-OBS-ABSA | **89.50** | **86.37** | **88.25** | **93.10** | **78.54** | **78.29** |

insights of BYD-OBS-ABSA. While CoT-GPT improves upon Vanilla GPT by incorporating a chain-of-thought prompting mechanism, it still falls short of leveraging comprehensive contextual insights. The reasoning process, though step-by-step, remains static and can propagate errors from initial steps without correction mechanisms.

In conclusion, our BYD-OBS-ABSA framework significantly outperforms both Vanilla GPT and CoT-GPT in few-shot settings across various datasets. The integration of emotion-oriented in-contexts and self-adaptive bootstrap optimization allows our model to utilize few-shot examples more effectively, leading to superior performance. These findings underscore the potential of our approach to address the challenges of sentiment analysis in low-resource settings and pave the way for future research to further optimize this framework for domain-specific data distributions and performance improvements.

## 5.9 | Case Studies

### 5.9.1 | SemEval-2014 Example in Table 4

Here, as shown in Table 4, we examine a typical instance from the SemEval-2014 dataset to illustrate the effectiveness of our BYD-OBS-ABSA framework in handling ABSA tasks compared to a Large Language Model (LLM) Chain-of-Thought (CoT) approach. The sentence under consideration is, "The Management was less than accomodating," with the aspect term identified as "Management". The gold label for this aspect is "negative".

In BYD-OBS-ABSA, we employ a multi-step reasoning process that integrates emotion-oriented constraints, contextual backgrounds, and analogical reasoning to determine sentiment. Initially, the framework evaluates whether the sentiment towards "Management" can be inferred as positive, negative, or neutral. In this case, the statement "less than accommodating" clearly indicates a negative sentiment, suggesting dissatisfaction or criticism. BYD-OBS-ABSA concludes that the sentiment conveyed by the statement is negative, indicating dissatisfaction with the management's level of accommodation. This negative inference is supported

| Original Annotation for Example 09 from **SemEval-2014**. |
|---|
| **Sentence:** The Management was less than accomodating. |
| **Aspect:** Management |
| **Gold Label:** negative |

| The LLM Reasoning Procedure of BYD-OBS-ABSA. |
|---|
| **# Constrain context:** |
| Let's analyze each question based on the given statement: 1. **Can we infer that the rose roll is positive?** - The phrase "less than accommodating" implies that the Management was not very accommodating, which suggests a negative or at least not positive impression. - *PRED*: No 2. **Can we infer that the rose roll is negative?** - The phrase "less than accommodating" conveys a negative sentiment towards the Management, implying dissatisfaction or disappointment. - *PRED*: Yes 3. **Can we infer that the rose roll is neutral?** - The term "less than accommodating" indicates a deviation from neutrality towards a negative perception, hence Management cannot be inferred as neutral based on this statement. - *PRED*: No |
| **# Background context:** |
| Management is the process of coordinating and overseeing resources, people, and tasks within an organization to achieve specific goals effectively and efficiently. It involves planning, organizing, leading, and controlling various activities to ensure that objectives are met in a structured manner. |
| **# Analogies context:** |
| 1. **Sentence:** "The hotel's management responded promptly to our complaints about noise, showing great concern for guest comfort." - **Predicted sentiment polarity:** 'Positive' - **Explanation:** The word "promptly" and the phrase "great concern for guest comfort" indicate efficient and caring management, likely leading to a positive sentiment. 2. **Sentence:** "The new management team has implemented several cost-cutting measures, which has led to decreased employee morale." - **Predicted sentiment polarity:** 'Negative' - **Explanation:** The phrase "decreased employee morale" suggests that the management's actions have had a negative impact, hence a negative sentiment is predicted. 3. **Sentence:** "The restaurant's management consistently receives praise for its friendly and attentive service." - **Predicted sentiment polarity:** 'Positive' - **Explanation:** The words "praise," "friendly," and "attentive" all indicate positive sentiments towards the management, suggesting good service and customer satisfaction. |
| **@@ GPT-3.5 Response:** |
| Given the provided analysis framework and contexts, from 'The Management was less than accomodating.', we can infer that the sentiment polarity towards the 'Management' (ASPECT) is **negative**. |
| @@ `function` sentiment_classification_metric(·) ⟶ Sentiment: **Negative** |

| The LLM Reasoning Procedure of LLM CoT. |
|---|
| **# LLM Chain-of-Thought (CoT) on GPT-3.5-turbo:** Given the sentence "The Management was less than accomodating." and the aspect "Management", determine the sentiment towards the aspect in the sentence. Please think step by step. |
| **@@ CoT Reasoning:** Sure! Let's analyze the sentiment towards the aspect "Management" in the sentence "The Management was less than accommodating." 1. **Identify the sentiment indicator**: The phrase "less than accommodating" is the sentiment indicator in the sentence. 2. **Interpret the sentiment indicator**: "Accommodating" typically implies a positive trait, suggesting willingness to help or adjust to meet needs. 3. **Apply the sentiment to the aspect**: Since "less than accommodating" suggests a shortfall in being helpful or flexible, I might incorrectly infer a negative sentiment towards the Management. 4. **Conclusion**: Based on the interpretation that "less than accommodating" implies negativity towards the Management, I would erroneously conclude that the sentiment towards "Management" in the sentence "The Management was less than accommodating" is negative. |
| **@@ Prediction** ⟶ Sentiment: **Negative** |

**TABLE 4** The typical prediction example from **SemEval-2014** where our method (BYD-OBS-ABSA) correctly predicts (Negative), while the LLM-based Chain-of-Thought on `GPT-3.5-turbo` makes an incorrect prediction (Positive).

| Original Annotation for Example 25 from **SemEval-2015**. |
|---|
| **Sentence:** This place has ruined me for neighborhood sushi. |
| **Aspect:** sushi |
| **Gold Label:** positive |

| The LLM Reasoning Procedure of BYD-OBS-ABSA. |
|---|
| **# Constrain context:** |
| Let's analyze each question based on the given statement: 1. **Can we infer that the sushi is positive?** - The statement "This place has ruined me for neighborhood sushi" implies that the sushi at "this place" is exceptionally good or high-quality compared to the neighborhood sushi. Therefore, we can infer that the sushi at "this place" is positive. - *PRED*: Yes 2. **Can we infer that the sushi is negative?** - The statement does not suggest anything negative about the sushi itself at "this place". It only suggests that the sushi is so good that it has raised the standard or expectation for sushi elsewhere. Hence, we cannot infer that the sushi is negative. - *PRED*: No 3. **Can we infer that the sushi is neutral?** - Again, the statement indicates a comparison where the sushi at "this place" is better than neighborhood sushi. It does not imply neutrality but rather a positive comparison. - *PRED*: No |
| **# Background context:** |
| Sushi is a quintessential Japanese dish that has gained immense popularity worldwide. It typically consists of vinegared rice (shari) combined with various ingredients such as raw or cooked seafood, vegetables, and sometimes tropical fruits. The most recognizable form is nigiri sushi, where a small mound of rice is topped with slices of fish or other toppings like shrimp or egg. Another well-known type is maki sushi, which involves rolling the rice and fillings inside a sheet of seaweed (nori) and slicing it into bite-sized pieces. |
| **# Analogies context:** |
| 1. **Sentence:** "The sushi at that new restaurant is absolutely delicious, especially the salmon rolls." - **Predicted sentiment polarity:** 'positive' - **Explanation:** The word 'delicious' indicates a positive sentiment towards the aspect 'sushi'. 2. **Sentence:** "I found the sushi place near the office to be quite average; nothing special." - **Predicted sentiment polarity:** 'neutral' - **Explanation:** The phrase 'quite average; nothing special' suggests a neutral sentiment, neither positive nor negative. 3. **Sentence:** "The sushi chef seemed inexperienced; the rolls were poorly made and lacked freshness." - **Predicted sentiment polarity:** 'negative' - **Explanation:** Words like 'inexperienced', 'poorly made', and 'lacked freshness' convey a negative sentiment towards the aspect 'sushi'. |
| **@@ GPT-3.5 Response:** |
| Given the provided analysis framework and contexts, from 'This place has ruined me for neighborhood sushi.', we can infer that the sentiment polarity towards the sushi (ASPECT) is **positive**. |
| @@ `function` sentiment_classification_metric(·) ⟶ Sentiment: **Positive** |

| The LLM Reasoning Procedure of LLM CoT. |
|---|
| **# LLM Chain-of-Thought (CoT) on GPT-3.5-turbo:** Given the sentence "This place has ruined me for neighborhood sushi." and the aspect "sushi", determine the sentiment towards the aspect in the sentence. Please think step by step. |
| **@@ CoT Reasoning:** This place has ruined me for neighborhood sushi. Aspect: sushi Reasoning: Letś think step by step in order to produce the sentiment. We can interpret from the phrase "ruined me for" that the speakerś experience with sushi at this particular place has set a high standard or spoiled their preference for sushi in their neighborhood. This suggests dissatisfaction or disappointment with the quality or experience of neighborhood sushi in contrast to the sushi from "this place". |
| **@@ Prediction** ⟶ Sentiment: **Negative** |

**TABLE 5** The typical prediction example from **SemEval-2015** where our method (BYD-OBS-ABSA) correctly predicts (Positive), while the LLM-based Chain-of-Thought on `GPT-3.5-turbo` makes an incorrect prediction (Negative).

by a comprehensive background context describing typical expectations of management behavior, enhancing the understanding of the aspect term.

Analogies drawn from similar sentences with negative sentiments towards management further solidify BYD-OBS-ABSA's prediction of a negative sentiment in this context. This approach contrasts with the reasoning of the LLM CoT using GPT-3.5-turbo, which might interpret the sentiment differently due to the complexity of implicit negative tones like "less than accommodating." The LLM CoT may not capture the nuanced dissatisfaction implied by the statement, focusing more on surface-level cues.

In summary, BYD-OBS-ABSA effectively analyzes sentiments like those expressed towards "Management" in the provided example, ensuring a nuanced understanding that aligns with human interpretation.

In another instance from **SemEval-2014**, e.g., the sentence, "The Management was less than accommodating," with the aspect term "Management". BYD-OBS-ABSA correctly identifies the sentiment as negative, while the CoT approach incorrectly predicts a positive sentiment. This error highlights the limitations of the CoT approach in capturing the negative connotation implied by the phrase "less than accommodating".

### 5.9.2 | SemEval-2015 Example in Table 5

As shown in Table 4, the second example from the SemEval-2015 dataset involves the sentence, "This place has ruined me for neighborhood sushi," with the aspect term "sushi". The gold label for this aspect is "positive". BYD-OBS-ABSA again demonstrates its superior reasoning capabilities by dissecting the sentiment conveyed in the statement.

BYD-OBS-ABSA's constrained context analysis deduces that the phrase "ruined me for neighborhood sushi" implies that the sushi at this place is of exceptionally high quality, surpassing that of neighborhood sushi. This inference is drawn from the comparison inherent in the statement, which elevates the sushi at this place by diminishing the appeal of neighborhood sushi. Consequently, the sentiment is inferred to be positive. The background context further enriches this analysis by providing a detailed description of sushi, including its various forms like nigiri and maki, which helps in understanding the high-quality connotation implied by the statement.

Analogies offer additional support, comparing sentences with explicitly positive, neutral, and negative sentiments towards sushi. These analogies reinforce the positive sentiment inferred by BYD-OBS-ABSA. On the other hand, the LLM CoT approach misinterprets the sentiment as negative, interpreting "ruined me for neighborhood sushi" as an expression of dissatisfaction with the neighborhood sushi rather than an endorsement of the high-quality sushi at this place.

In another instance from **SemEval-2015**, e.g., the sentence, "The space is limited so be prepared to wait up to 45 minutes - 1 hour, but be richly rewarded when you savor the delicious indo-chinese food," with the aspect term "space". BYD-OBS-ABSA correctly identifies the sentiment as positive, while the CoT approach incorrectly predicts a neutral sentiment. This error highlights the limitations of the CoT approach in capturing the negative connotation implied by the phrase "be richly rewarded".

In conclusion, these case studies highlight the robustness and accuracy of BYD-OBS-ABSA in performing ABSA tasks. By incorporating emotion-oriented in-context constraints, backgrounds, and analogical reasoning, our framework surpasses the LLM CoT approach, providing more accurate sentiment predictions. This demonstrates the effectiveness of integrating comprehensive contextual insights into LLM-based ABSA, addressing the challenges of LLM hallucinations and improving overall sentiment analysis performance.

## 6 | DISCUSSIONS

### 6.1 | Discussion on BYD-OBS-ABSA in FSL and ZSL

- **BYD-OBS-ABSA in Few-Shot Learning (FSL):** BYD-OBS-ABSA effectively leverages the few-shot learning paradigm by integrating various emotion-oriented contextual techniques. These techniques provide the model with rich emotional and contextual cues, significantly enhancing its ability to generalize from limited examples. The framework's capability to adapt to new tasks with minimal data demonstrates its robustness and flexibility, marking a substantial improvement over traditional small language models (SLMs).
- **BYD-OBS-ABSA in Zero-Shot Learning (ZSL):** Even in zero-shot learning scenarios, where no specific training data is available for the target task, BYD-OBS-ABSA exhibits exceptional performance. By employing sophisticated prompt engineering techniques, including Chain-of-Thought (CoT) and In-Context Learning (ICL), the framework effectively utilizes pre-trained knowledge. The inclusion of emotion-oriented contexts ensures that the model maintains high accuracy and contextual understanding, even without direct exposure to the target domain.

### 6.2 | Limitations of BYD-OBS-ABSA

While BYD-OBS-ABSA significantly enhances ABSA tasks by leveraging large language models and emotion-oriented contexts, it is not without limitations. The framework's reliance on large pre-trained language models may introduce challenges related to computational resources and efficiency, given the multiple rounds of LLM interactions required to solve a single ABSA task. Additionally, the prediction performance of BYD-OBS-ABSA can be subject to random fluctuations due to the inherent variability in LLM responses, although these fluctuations are limited and do not hinder the overall sentiment prediction performance. Ensuring the consistency and relevance of contextual data is crucial for maintaining the model's robustness and accuracy across different domains and applications.

### 6.3 | Implications of Future Research

BYD-OBS-ABSA offers substantial potential for future research and practical applications in sentiment analysis and natural language processing. Its ability to handle few-shot and zero-shot learning scenarios with high accuracy opens new avenues for developing robust sentiment analysis models in low-resource settings. In real-world applications, the framework can be utilized for fine-grained sentiment analysis in customer reviews, social media monitoring, and market research, providing deeper insights into user opinions and enhancing decision-making processes.

Future research can explore extending the BYD-OBS-ABSA framework to support multilingual and cross-domain sentiment analysis. Integrating additional modalities, such as images and videos, could further enhance the model's capabilities, enabling comprehensive sentiment analysis across diverse data sources. Addressing these challenges will be crucial for developing more versatile and effective sentiment analysis systems.

### 6.4 | Ethical Considerations

thical considerations are paramount in developing the BYD-OBS-ABSA framework. The ability to accurately analyze sentiments and understand contextual nuances can significantly impact decision-making processes and public opinion. Therefore, it is essential to ensure that the framework is used responsibly and ethically, avoiding potential misuse in areas such as biased sentiment analysis or manipulation of public opinion. Adhering to ethical guidelines and promoting transparency in the deployment of sentiment analysis models will be critical in maintaining the integrity and trustworthiness of these technologies.

## 7 | CONCLUSION AND PERSPECTIVE

In this study, we introduced the BYD-OBS-ABSA framework, an innovative approach designed to enhance aspect-based sentiment analysis (ABSA) using large language models (LLMs). Our framework integrates emotion-oriented in-context constraints, backgrounds, and analogical reasoning to address the limitations of traditional LLM prompting techniques and mitigate the impact of LLM hallucinations. By leveraging sophisticated in-context learning strategies and self-adaptive bootstrap instructions optimization, BYD-OBS-ABSA demonstrates superior performance in accurately determining sentiment polarity for aspect terms. Through extensive experiments with `GPT-3.5-turbo` across six public datasets, we validated the effectiveness and robustness of our framework. This highlights the potential of our approach to provide more reliable and nuanced sentiment insights, which are crucial for applications in customer service, marketing, and product development. In the future, we will focus on refining BYD-OBS-ABSA to further improve its scalability and adaptability to different LLM backbones, domains, and languages. Additionally, exploring the integration of more advanced in-context learning methods and fine-tuning strategies will be essential to push the boundaries of LLM capabilities in sentiment analysis. Our open-source code repository on GitHub[11] is available to facilitate further research and development in the field of linguistic processing.

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## conflict of interest

There is no conflict of interest to declare.

## data availability statement

The data and implementation codes that support the findings of this study are publicly available on GitHub: https://github.com/albert-jin/BYD-OBS-ABSA.

## endnotes

[1] Our Code Repository of BYD-OBS-ABSA: https://github.com/albert-jin/BYD-OBS-ABSA.
[2] DSPy tutorial: https://dspy-docs.vercel.app/docs/intro
[3] The task details can be accessed at http://alt.qcri.org/semeval2014/task4/.
[4] SemEval-2015 is available at https://github.com/lixin4ever/BERT-E2E-ABSA/tree/master/data.
[5] SemEval-2016 is available at https://github.com/lixin4ever/BERT-E2E-ABSA/tree/master/data.
[6] Twitter is available at https://github.com/lixin4ever/E2E-TBSA/tree/master/data.
[7] Dataset is available at https://github.com/songyouwei/ABSA-PyTorch/tree/master/datasets/acl-14-short-data.
[8] BERT: https://huggingface.co/bert-base-uncased.
[9] The Code Running Environment of BYD-OBS-ABSA: https://colab.research.google.com/.
[10] https://github.com/albert-jin/BYD-OBS-ABSA
[11] Github of BYD-OBS-ABSA: https://github.com/albert-jin/BYD-OBS-ABSA.

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